MILITARY SPECIFICATION

TEST SET, AIRBORNE DATA TERMINAL

This Specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for design, construction, performance and testing of the test set, Air Data Terminal, referred to herein as the equipment.

1.2 Associated equipment. This equipment should operate with the associated equipment listed in 6.6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation - Packaging Methods of
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic and Associated Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted (ASG)
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Requirement for
MIL-M-38510	Microcircuit, General Specifications For
MIL-H-46855	Human Engineering, Requirements for Military Systems, Equipment and Facilities
MIL-C-81774	Control Panel, Aircraft, General Requirement for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

NAVAL AIR SYSTEMS COMMAND

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
AS-4613	Application and Derating Requirements for Electronic Components, General Specification for
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliabiity Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the currentversion before use.

MIL-STD-790	Reliability Assurance Program for Electronic Parts, Specifications
MIL-STD-794	Parts and Equipments Procedures, Packaging and Packing
MIL-STD-810	Environmental Test Methods
MIL-STD-882	System Safety Program Requirements
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	List of Standard Microcircuits
MIL-STD-2068	Reliability Development Tests
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2084	General Requirements for Maintainability of Avionic and Electronic Systems and Equipment

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

Other Government documents, drawings, and publications. The following other Government documents and drawings form a part of this specification to the extent specified herein.

DARCOM-P-706-410	Engineering Design Handbook for Electromagnetic Compatibility

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified, a sample(s) shall be subjected to First Article Inspection (see 4.4 and 6.4).

3.2 Parts and Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic Technology shall be considered and microelectronic items shall conform to the requirements specified herein.

b. All parts and materials shall conform to the applicable requirements of MIL-T-28800.

c. Nonrepairable subassemblies shall be used to conform to MIL-STD-2084 and as defined in MIL-T-28800.

d. All microelectronic devices and parts shall be readily available, off-the-shelf from at least two qualified sources. All second source microelectronic devices and equipment shall be mechanically and electrically interchangeable.

e. Circuits shall be designed to use a minimum of different materials, parts, subassemblies, cards and modules.

3.2.1 Nonstandard Parts and Materials Approval. Approval for the use of non-standard parts and materials shall be obtained to conform with MIL-E-5400. Microelectronics Devices shall be approved to conform with MIL-STD-965 except that approval as a non-standard part shall not be required for devices which comply with all the following: (See 6.2.2)

a. Listed in MIL-STD-1562;

b. Available from at least two sources which are listed in the appropriate QPL of MIL-M-38510, and

c. Procured to conform to the requirements of MIL-M-38510 from a qualified source.

3.2.2 Microelectronic modular assemblies. When used, Microelectronic Module Assemblies shall conform to MIL-STD-454, Requirement 64 and MIL-STD-883, Class B. Conformal coatings, encapsulants, enbedments or potting materials used with modular assemblies containing integrated circuits and descrete parts shall be easily removable without damage to the assembly, with the exception of the Voltage Controlled Oscillator in the R.F. Generator Module.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the current version before use.

3.2.3 Modules. The electronic portions of the equipment shall be functionally modularized to conform to MIL-STD-2084.

3.2.4 Electronic Parts Screening and Derating.

3.2.4.1 Electronic Parts Screening.

3.2.4.2 Electronic parts derating. The contractor shall use MIL-HDBK-217 as the basis for derating of the parts. This derating shall encompass the appropriate and meaningful application conditions such as voltages, current, power, temperature, mechanical and duty cycles. This deration shall be at least as restrictive as specified in AS-4613. The contractor shall conduct a part level stress analysis to verify derating compliance. This analysis shall verify all parameter deratings are compliant under the following conditions.

a. Nominal assembly or subassembly power dissipation.

b. Worst-case piece part stress.

c. Worst-case equipment thermal environment.

This analysis shall be part of a Part Level Stress Analysis and Prediction Report. Any noncompliance to the derating criteria shall require approval by the procuring activity prior to final design acceptance. The following types of parts shall require the approval of the procuring activity prior to being used in new designs:

a. Hot carrier (Schottky) rectifiers

b. PROMS

c. Wet (sintered slug) tantalum Capacitors for use other than on Voltage Regulator Assemblies and low leakage current requirements.

d. Plastic Epoxy or other encapsulated semiconductors.

e. Germanium semiconductors

f. Aluminum electrolytic capacitors

g. Transformers and inductors of open construction configuration

3.3 Design and construction. The equipment shall conform to the applicable requirements of MIL-T-28800 for design, construction and workmanship, except as otherwise specified herein. Solid state design shall be employed throughout.

3.3.1 Total weight and dimensions

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.1.1 Total weight. The total weight of the equipment including the enclosure shall be a minimum consistent with good design and shall be not greater than 72 pounds.

3.3.1.2 Dimensions. The equipment shall be enclosed in a case, style A to conform to MIL-T-28800 whose dimensions shall be not greater than 20 inches wide by 18 inches high, by 28 inches long. The maximum dimensions shall include handles in extended position and any projections provided for stacking.

3.3.1.3 Color. The exterior color of the equipment shall be color V yellow (gloss), No. 13538 of FED-STD-595). The color of the control panel shall be color T Black (Lusterless), (No. 37038 of FED-STD-595).

3.3.2 Reliability. The contractor shall conduct a reliability program utilizing MIL-STD-785 as a guide. (See 6.2.2)

3.3.2.1 Operational stability. The equipment shall operate with specified performance, continuously or intermittently, for a period of at least 900 hours in the operational environment without the necessity for readjustment of any controls which are inaccessable to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have an operating life of not less than 15,000 hours with minimal servicing and replacement of parts.

3.3.2.3 Storage life. The equipment shall be designed to meet the performance and service life requirements of the specification after five years storage under worldwide climatic conditions specified in MIL-STD-210 when packaged to conform with MIL-STD-794 Level A and MIL-P-116 Method 11A.

3.3.2.4 Reliability in Meantime Between Failure (MTBF). The equipment including any Built-In-Test (BIT) provision shall have an upper test MTBF (θ_0) for continuous and intermittent operation, of 900 hours and a lower test MTBF, (θ_1) of 450 hours when tested and accepted as specified under the requirements of 4.4.3.

3.3.2.5 Time totalizing meter. The equipment shall have a time totalizing meter to conform to MIL-M-7793. The meter shall be located such that it may be read with the removal of the top cover. The meter shall indicate "ON" time which is any operating mode including standby and shall have a range of 0 to 9,999 hours and shall not be capable of being reset.

3.3.2.6 Reliability program specific requirements. To prepare the necessary program consistency and program control visibility, the specific requirements specified in 3.3.2.6.1 through 3.3.2.6.1.4 shall apply to contract reliability programs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.2.6.1 Reliability Design Analysis. The contractor shall perform detailed reliability design analysis to conform to MIL-STD-785. (See 6.2.2). The analysis shall consider failure modes and effects, component loadings, environments, and failure rates, and mean-time-between-failures. Analysis shall continue throughout the program design, development, and test phases and shall progressively refine reliability predictions with the most current available design and environmental data. In general, early reliability projections shall be based upon estimated system configuration, component part ratings and stresses and environmental conditions. Later predictions shall replace these estimates with actual design details, calculated loading and calculated local environmental exposures. Predictions in the final phase of the program shall reflect current design (i.e. include the latest design changes) and the component loadings and environmental exposures observed in the development test, adjusted where necessary to reflect maximum specified equipment ambients.

3.3.2.6.1.1 Failure modes effects and criticality analysis. The contractor shall prepare a reliability analysis which shall include all significant failure modes. The analysis shall document failure modes, effects and critical analysis. The failure mode effects and criticality analysis shall be documented in sufficient detail to enable correction of each distinct failure mode having significant impact on specified equipment performance. (See 6.2.2)

3.3.2.6.1.2 Environment. Component failure rate predictions shall reflect the equipment's maximum specified environmental exposures, as defined in this specification and the estimated and test observed, local component part environments present within the equipment during operation at the specified equipment ambients. Predictions shall acknowledge heat generated within the black box and the resulting local components temperature rises above the maximum specified black box ambient. Prediction shall also consider aircraft interface and operation and the resulting usual increase in failure rates above those observed in laboratory test. Component failure rate predictions, based upon MIL-HDBK-217, shall reflect the handbook's adjustment factors. Predictions based on other data shall be similarly adjusted as appropriate.

3.3.2.6.1.3 Statistics. Analysis of electronic components with wearout characteristics and age-dependent failure rates (e.g., lamps switches, etc.), and mechanical components shall consider the statistical implementations of this wearout. Analysis may neglect only the wearout precluded by the scheduled maintenance procedures as defined in the hardware specification.

3.3.2.6.1.4 MTBF Predictions. The reliability design analysis shall provide predictions of equipment and system MTBF. MTBF shall be based on all component parts in a series probability model. MTBF shall consider all redundancies which permit operation in accordance with specified requirements, despite component failure modes, and redundancies. This shall include the failure sensing elements and the mechanisms used to switch from a failed path to an alternate path. Analysis shall also consider testing done to insure all redundant paths are operable at the start of a mission. Degraded operation modes associated with redundancies shall be documented. It shall also consider component stress level variations with mode of operation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.2.6.2 Failure analysis. The contractor shall prepare a failure analysis and corrective action report of all equipment failures occurring during: (See 6.2.2)

a. Contract tests specified by the equipment specification.

b. Service use during the contractor support phase of the program.

Contractors shall employ failure analysis capabilities and facilities of the type necessary to derine the cause of failure, including those described in MIL-STD-790. Positive correction action shall be implemented to prevent recurrence of all failures. Program planning in response to MIL-STD-785, and Management Tasks, shall include the necessary procedures for data collection and failed specimen control and routing.

3.3.2.6.2.1 Test data. The contract shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. (See 6.2.2). In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following: (See 4.6.5)

a. Part type number
b. The circuit reference symbol number
c. The part function
d. Name of the manufacturer
e. Nature of the failure
f. The number of hours which the part operated prior to failure

3.3.3 Cabling and connections

3.3.3.1 Cables and connectors. The equipment shall provide for the use of cable and connectors to conform to MIL-T-28800.

3.3.3.2 Interconnection cabling. The equipment shall be capable of specified operation using external wiring to conform to the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cables to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall be supplied as part of the equipment.

3.3.4 Control panels. The equipment control panel shall conform to the applicable requirements of MIL-C-6781, MIL-STD-1472 and MIL-T-28800. Controls and indicators on the control panel shall be grouped by test set function. The configuration of the panel shall be approved by the procuring activity prior to prototype fabrication. (See 6.2.2). The control panel shall be a Shop Replaceable Assembly (SRA).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.4.1 Controls. The equipment controls shall conform to the applicable requirements of MIL-T-28800, MIL-STD-454, Requirement 28 and MIL-STD-1472.

3.3.4.2 Rotary switches. In addition to the requirements of MIL-T-28800 and MIL-STD-1472, rotary switches shall align for each position with the front panel index marks such that no confusion can exist as to swtich setting.

3.3.5 Interchangeability. Interchangeability shall conform to MIL-STD-454, Requirement 7.

3.3.6 Electromagnetic interference (EMI). The generation of electromagnetic interference by the equipment, and the vulnerability of the equipment to electromagnetic interference shall conform to the requirements of MIL-STD-461 for Class A1 equipment except as modified herein. The equipment shall be tested in accordance with MIL-STD-462, Notice 1 and 2.

3.3.6.1 Design guidelines. The equipment shall conform to the requirements of AD-1115 and DARCOM-P-706-410 in addition to MIL-STD-461.

3.3.6.2 Emission limits. The following MIL-STD-461, Part 2 limits apply.

3.3.6.2.1 CE01

3.3.6.2.2 CE03

3.3.6.2.3 CE07

3.3.6.2.4 RE01

3.3.6.2.5 RE02

3.3.6.3 Susceptibility limits. The following MIL-STD-461, Part 2 limits apply as modified:

3.3.6.3.1 CS01

3.3.6.3.2 CS02 except that this requirement shall apply to all neutrals and ground whether or not they are internally returned to chassis ground.

3.3.6.3.3 CS03

3.3.6.3.4 CS04

3.3.6.3.5 CS05

3.3.6.3.6 CS06

3.3.6.3.7 RS02

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.6.3.8 RS03. The field levels and modulation shall be as specified in Table I, RS03 Signals, and Table II, High Level RS03 Signals. The susceptibility criteria of 3.3.6.4.2 does not apply to high level RS03 testing.

3.3.6.4 Susceptibility. The following criteria shall be applied to determine whether the display is susceptible.

a. Any repeatable, non-commanded change in the control settings, panel settings, modes of operation, etc., of the equipment shall be considered an indication of susceptibility.

(1) This shall apply for both transient and permanent changes.

(2) The inability to change modes of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility.

b. The equipment shall conform to the performance requirements of 3.5.

3.3.6.5 Grounding requirements. The grounding system between SRAs of the equipment shall be compatible and consistent with the following equipment grounding scheme. All SRAs shall be grounded in such a manner as to prevent ground loops in common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.3.6.5.1 Chassis grounds. A ground wire of minimum length connected internally to the equipment shall be provided at a pin on the primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.3.6.5.2 Shield grounds. A separate connector pin shall be provided for each external wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis adjacent to the connector mounting and by the shortest means practical.

3.3.6.5.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, bushings, etc., shall be grounded to the chassis.

3.3.6.6 Bonding. A means of electrically grounding the equipment to the aircraft chassis shall be provided. The grounding circuit resistance shall be not greater than 2.5 milliohms. Grounding hardware shall be compatible with the SH-60B Grounding Provisions.

3.3.6.7 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. RSO3 Signals

Frequency (MHz)	Modulation		Power Density (mW/cm)[5]		v/m(Ref)[6]
			Peak[5]	Avg	
0.014-850	AM 90% 1 kHz Sine Wave		-	.106	20
215-225[7]	PAM[1] 10 usec[2] PW[3]	200 pps[4]	.317	-	35
215-225	PAM 200 usec PW	200 pps	.317	-	35
400-450	PAM 15 usec PW	300 pps	.106	-	20
850-940[7]	PAM 2 usec PW	1000 pps	.106	-	20
850-940	PAM 125 usec PW	300 pps	.106	-	20
940-1215	PAM 10 usec PW	1000 pps	.106	-	20
1215-1365	PAM 10 usec PW	3000 pps	.106	-	20
1365-2900	PAM 1 usec PW	1000 pps	.106	-	20
2900-3100[7]	PAM 3 usec PW	1000 pps	1.6	-	78
2900-3100	PAM 70 usec PW	200 pps	1.6	-	78
3100-9000	PAM 1 usec PW	1000 pps	.106	-	20
9000-14000	PAM 0.2 usec PW	1000 pps	.106	-	20
14000-18000	PAM 0.2 usec PW	50000 pps	.106	-	20

1/ Pulse Amplitude Modulated.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; i.e., average power density shown for reference only.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and/or field strength shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE II. High Level RSO3 Signals

Frequency (MHz)	Modulation	Power Density (mW/cm^2) 5/	v/m (ref) 6/	SCAN RATE (sec) 7/	DWELL TIME (msec) 7/
200	PAM 1/ 10 usec 2/ PW 3/ 200 pps 4/	10.6	200	4	500
220	PAM 200 usec PW 200 pps	10.6	200	4	500
425	PAM 15 usec PW 300 pps	10.6	200	10	200
900 8/	PAM 2 usec PW 1000 pps	20	275	5	50
900	PAM 125 usec PW 300 pps	20	275	5	50
1250	PAM 10 usec PW 3000 pps	10.6	200	5	50
1350	PAM 5 usec PW 3000 pps	10.6	200	5	50
3000 8/	PAM 3 usec PW 1000 pps	160	775	5	50
3000	PAM 70 usec PW 200 pps	160	775	5	50
3600	PAM 1 usec PW 1000 pps	100	610	5	50
5600	PAM 1 usec PW 600 pps	10.6	200	5	50
9100	PAM 0.5 usec PW 1000 pps	10.6	200	5	50
9375	PAM 0.2 usec PW 1000 pps	10.6	200	5	50
13900	PAM 0.2 usec PW 50000 pps	10.6	200	0	CONT
15500	PAM 0.2 usec PW 50000 pps	10.6	200	0	CONT
33200	PAM 0.2 usec PW 2000 pps	0.106	20	0	CONT

1/ Pulse Amplitude Modulated.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; i.e., average power density shown for reference only.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disenabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and/or field strength shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.7 Maintainability. The equipment shall have a specified mean corrective maintenance time (MCT) of two hours and a maximum corrective maintenance time (MCT max) of four hours at the intermediate and depot level. The equipment MCT includes localization, isolation, disassembly, interchange, reassembly, alignment, and test. The equipment shall be designed such that no periodic calibration is required.

3.3.7.1 Maintainability philosophy. To provide for increased system effectiveness, maintainability shall receive equal consideration with such factors as cost, physical parameters and reliability. The maintenance concept to be implemented in the maintainability program is to perform corrective maintenance by utilizing manually initiated built-in-test for fault isolation to the appropriate SRA or assembly which will then be removed and replaced at the intermediate level. The defective SRA or assembly will be forwarded to the depot maintenance level for repair.

3.3.7.2 Maintainability Program. The contractor shall establish and conduct a maintainability program to conform to MIL-STD-470. (See 6.2.2). All intermediate maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

a. Civilian Education: Ninth grade education and reading level or equivalent.

b. Technical Training: Navy Class A and Class C technical schools appropriate to the maintenance task.

c. Experience: One year technical experience in addition to technical training.

3.3.7.3 Provisions for maintainability. The maintainability program built-in-test program built-in-test features, construction and packaging provisions for test points and other maintainability parameters shall conform to MIL-STD-2084 except as modified herein. References to system BIT, BITE, and ORT requirements in MIL-STD-2084 shall be interpreted as applying to the equipment as a stand alone system.

3.3.7.4 Packaging. The equipment shall be packaged in modular form. All electrical or electronic circuits and parts shall be packaged as replaceable and repairable plug-in modules and easily removable and replaceable at the intermediate level. Circuit breakers, indicators, and other items which may logically be bolted, or otherwise more rigidly electrically and mechanically fastened to the equipment casing shall not be considered modules. Non module items shall be kept to an absolute minimum. All parts and assemblies which the contractor desires to install as non-module items shall require the prior written approval of the procuring activity. Special tools shall not be required for module removal. In addition, the circuits shall be designed so that a defective functional loop can be isolated to a specific SRA or assembly.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.7.4.1 Module commonality. The number of different types of modules shall be kept to a minimum to ease the testing and spares requirements at all levels of maintenance. Unused functions on a module are permitted if commonality is improved.

3.3.7.4.2 Module interchangeability. Modules which perform identical functions shall be identical and interchangable throughout the equipment and to the extent possible with the AN/ARQ-44 and AN/SRQ-4 equipments. This requirement takes precedence over 3.2d.

3.3.7.4.3 Module access. Each module shall be removed and replaced at the intermediate level. Access to any module in a power-off state shall be achieved by the removal of at most one protective cover or enclosure exclusive of the equipment case. Access to all modules indicated as faulty suspects in any single automatic fault isolation outcome shall be achieved by the removal of, at most, one protective cover or enclosure. Each module or hardware subunit shall be capable of being removed and replaced without prior removal or replacement of any other module or hardware subunit. Each module shall have guides for insertion and removal which prevent accidental damage to adjacent modules, and connectors.

3.3.7.4.4 Module retention. Equipment designed to use screws or other similar devices for module retention shall be so designed that the retention devices are made captive to the module or equipment, with the exception of the RF Modules.

3.3.7.4.5 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or module into an incorrect location or position within or on the equipment.

3.3.7.4.6 Part replacement. Components and assemblies designated for intermediate level removal shall be designed to be replaced without disassembly and without disturbing elements of adjacent installations. The replacement of displays, display indicators and lamps shall accomplished without the requirement for equipment disassembly or the use of special tools.

3.3.7.4.7 Part replacement within modules. Part replacement within electronic modules shall be possible by skilled technicians working in a designated service area such as an intermediate maintenance activity. Excessive use of hard curing compounds (such as some epoxies) to hold parts in place is prohibited. Such compounds shall not be used on the wiring side of printed wiring boards. Hard curing compounds may be used sparingly when mechanical stability is essential to circuit performance. Conformal coatings conforming with MIL-STD-275 are not considered to be in conflict with this paragraph.

3.3.7.4.8 Nonrepairable subassemblies. Nonrepairable subassemblies shall be considered as nonstandard parts. All designs for nonrepairable assemblies will be approved by the procuring activity prior to release for fabrication. (See 6.2.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.7.4.9 Intermediate maintenance adjustments. Units (SRAs) or modules compirsing a system-sub-system shall not require any adjustment after a single unit or module has been replaced, except for RF output level calibration when components or modules within the RF, subsystem are replaced. The RF output calibration shall be performed with a single adjustment, requiring no sepcial tools.

3.3.7.4.10 Retention devices. Mechanical retention devices (latches) used for securing the equipment shall conform to MIL-T-28800 for Style A enclosures.

3.3.7.4.11 Test points. Test points shall be provided for intermediate and depot level maintenance to conform to MIL-STD-2084 and MIL-T-28800. Test points shall be compatible with the test equipment designated and planned for fleet use.

3.3.7.4.12 Compatibility with automatic test equipment (ATE). Compatibility with ATE is not required. The equipment shall conform to the maintenance philosphy as specified herein.

3.3.7.4.13 Adhesives. In addition to MIL-STD-454, Requirement 23 the use of adhesives shall result in a durable bond between the surfaces to which they are applied.

3.3.7.5 Built-in-test (BIT). The equipment shall provide for a start up BIT which gives a go/no-go indication to at least a 90 percent confidence level. Fault isolation to the SRA/Assembly exclusive of the Reed-Soloman code printed wiring assembly shall be as follows.

a. Three SRAs/Assemblies - 96 percent

b. Two SRAs/Assemblies - 93 percent

c. One SRA/Assembly - 85 percent

The indication shall be in the form of a test indicator to provide a qualitative means of checking the equipment operation and isolation of any fault or observed malfunction to a defective SRA. BIT isolation to faulty front panel switches, meters, and connectors, shall not be required. The front panel SRA shall be excluded from this requirement. Built-in-test indicator design shall be approved by the procuring activity prior to prototype fabrication.

3.3.7.5.1 Fail safe operation. The circuits and devices which provide the BIT function shall be designed to be fail safe and non-interactive such that any failure of the BIT will not affect or degrade the equipment performance in any manner.

3.3.7.5.2 Indicators. Indicators shall conform to MIL-STD-1472. A lamp test feature shall be provided to test all incandesent and light emitting semiconductor indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.7.5.2.1 GO/NO GO indicators. If latching type indicator(s) are used they shall be resettable without disassembly of the equipment and without special tools.

3.3.7.5.3 Readouts. Readouts shall conform to the requirements of MIL-STD-454, Requirement 68 and MIL-STD-1472.

3.3.8 Nomenclature, nameplates, and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall conform with MIL-N-18307. (See 6.2.2)

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

a.	Temperature	Room ambient (25°C ± 10°C)
b.	Altitude	Normal Ground (0-10, 000 ft.)
c.	Vibration	None
d.	Humidity	Room ambient up to 95% relative humidity
e.	Input power voltage	108-120 VAC 60-400 Hz Phase 1 five VA

3.3.10 Service conditions. The equipment shall operate as specified under any of the environemental service conditions or reasonable combination of these conditions to conform with MIL-T-28800 for Type I Class 3 equipment except as modified herein.

3.3.10.1 Vibration. The equipment shall operate as specified when subjected to the vibration conditions specified in MIL-T-28800 for Class 3 test equipment.

3.3.10.1.1 Bounce. The equipment shall operate as specified after being subjected to the loose cargo bounce test of MIL-STD-810, Method 514, Procedure XI, Part 2.

3.3.10.2 Temperature range. The equipment shall be designed to operate in the range of temperatures between -10°C to +55°C.

3.3.10.2.1 Overtemperature shutoff. Overtemperature protection circuits shall be provided which shall interrupt input power under conditions of excessive internal temperature. The power shall automatically reset when the temperature returns to a safe operating level. A latching type indicator should indicate to the operator that an over temperature condition has occured.

3.3.10.3 Humidity. The equipment shall withstand the effects of humidity up to 95 percent during operating and nonoperating conditions. During test the temperature shall be varied from 28 ± 10 degrees C to 55 degrees C. Length of exposure shall be 50 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.11 Warm up time. The time required for the equipment to warm up prior to operation in accordance with the specified performance requirements shall be kept to a minimum and shall be not more than three minutes under standard conditions and five minutes at extreme service conditions.

3.3.12 Input electrical power

3.3.12.1 Operating power. The equipment shall conform to MIL-T-28800 for Type I equipment. Operation from a DC power source or from battery is not required.

3.3.12.2 Loss of power. After accidental or deliberate stoppage of electrical power to the equipment regardless of the time in its operating cycle that such a stoppage should occur, and regardless of the time interval that elapses before power is restored, the equipment shall return to operation in its initialized state.

3.3.12.3 Abnormal power supply operation. The equipment power supply design shall provide for protection against any abnormal operation of any power supply in the equipment including undervoltage, overvoltage, overload and short circuit conditions. The protection design within the equipment be such that any failure mode of any power supply does not cause any component failure or chain reaction failures within the equipment. Fuses shall not be used. When all voltage levels return to normal, the equipment shall assume an operational state, ready to resume normal operation.

3.3.12.4 Circuit breakers. Circuit breakers shall be used in lieu of fuses and shall conform to MIL-STD-454, Requirement 37. The circuit breaker protection for the equipment may be intregal to the equipment on/off switch. The circuit breakers shall be accessible on the front panel of the equipment.

3.3.13 Cooling

3.3.13.1 Cooling provisions. The equipment shall be designed to provide for internal cooling. No external forced air cooling shall be required.

3.3.13.2 Thermal design. Thermal design shall conform to MIL-T-28800 and MIL-STD-454, Requirement 52.

3.3.14 Human performance/human engineering. Human performance and human engineering design criteria for the equipment shall conform to MIL-STD-1472. Applicable portions of MIL-C-81774, MIL-P-7788 and MIL-H-46855 shall be used as a guide for the contractors program.

3.3.15 Hazardous conditions. Under no circumstances shall the equipment be permitted to operate to the point at which it represents a safety hazard to surrounding equipment, personnel, or aircraft.

3.3.16 Soldering. The soldering of printed circuit wiring boards shall conform to the procedures and requirements of WS-6536.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

3.3.17 Safety. A system safety program shall be established to conform to MIL-STD-882. (See 6.2.2). In the design of the equipment, safety protection shall be provided, but be not limited to, the hazards of MIL- T -28800 for Type 1 equipment.

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and input power conditions.

3.4.1 Operation. The equipment shall provide either an RF or baseband interface with the AN/ARQ-44 and shall be capable of providing an uplink and receiving downlink signals from the AN/ARQ-44 as specified in 3.5. The equipment shall be capable of aiding the operator to fault isolate data link failures to the AN/ARQ-44 or to its interfaces. The equipment shall not provide any encryption capability.

3.5 Detail requirements. The equipment shall meet the requirements specified in 3.5.1 through 3.5.1.2.3.

3.5.1 Function. The equipment shall be capable of aiding the operator to fault isolate an uplink/downlink failure to an AN/ARQ-44 data link function or to the following system interfaces.

a. Receiver Group, Sonobuoy, AN/ARR-75

b. Signal Data Converter Group OU-103A (Radar Interface)

c. 1553A Data Bus

d. Communications System Control Group, CV 3519/ASC

3.5.1.1 Input signal characteristics and data formats. The equipment shall accept and respond to the following input signals and data formats.

3.5.1.1.1 Input RF signal format. The equipment shall receive G-band PM PSK modulated RF energy from the AN/ARQ-44 test port. Input RF characteristics shall be as follows:

a. The input RF signal shall be capable of withstanding +30 dBm at the RF input connector.

b. The test set RF input impedance shall be 50 ohms with a maximum VSWR of 1.5:1.

3.5.1.1.2 Input baseband signal characteristics. The input baseband signal shall be a ±1 V ± 10 percent signal for a characteristic impedance of 91 ohms ±5 percent single ended.

3.5.1.1.3 Input data format. The input data format shall be a multiplexed digital data stream consisting of voice, computer, discrete, and acoustic

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

words (ASW mode) or voice, computer, discrete and radar words (ASST mode). The equipment shall accept data formated so as to be compatible with AN/ARQ-44 downlink multiplex scheme.

3.5.1.2 Output signal characteristics and data formats.

3.5.1.2.1 Output RF signal characteristics. The equipment shall generate G-band FM FSK modulated RF signals which shall be supplied to the AN/ARQ-44 data terminal RF test port. The output RF signal characteristics shall be as follows:

a. Characteric impedance 50 ohms.

b. The RF level shall be +10 dBm ±1 dB unattenuated, for a VSWR not exceeding 1.5 to 1.

c. The test set shall not be damaged when the RF output port is either shorted or open.

3.5.1.2.2 Output baseband signal characteristics. The output baseband signal shall be a ±1 V ± 10 percent digital signal having a characteristic impedance of 91 ohms ±5 percent single ended.

3.5.1.2.3 Output data format. The output data format shall be a multiplexed data stream consisting of voice, computer, and discrete words. The test set shall generate data formated to be compatible with the AN/ARQ-44 uplink multiplex scheme.

3.5.2 Specific functions. The equipment shall provide the capability to selectively capture and display downlink computer messages. Messages shall be selected in accordance with their numerical identification. The equipment shall be capable of displaying individual computer data words as selected by the operator. The contents of these selected 16 bit data words shall be displayed to the operator in a sixteen bit binary format display. The contents of each message shall also be displayed to the operator using an alpha numeric display which shall identify the contents of each 16 bit data word specific to individual data fields. The equipment shall provide the following specific functions.

3.5.2.1 Computer down link message display. The equipment shall include in a readily available form (at a minimum plastic encapsulated paper) the binary definition of the computer word structure for the LAMPS MK III AOP to SPOP.

3.5.2.2 Acoustic multiplexed channel outputs. The equipment shall provide the capability to output the acoustic signal from any one of the eight downlink acoustic channels. Digital to Analog (D-A) conversion shall be such that the output shall be scaled from 0-16 Volts RMS. The acoustic output shall be capable of directly interfacing with the AN/UYS-1 Acoustic Detecting Set. The equipment shall include cables to provide an interface between the selected acoustic demultiplexed channel output and the aircraft acoustic

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the currentversion before use.

processor input. The interface shall be accomplished at wiring connector J-16 on the AN/USY-1 Acoustic Detecting Set. Phase linearity for the acoustic shall meet the linearity requirements for the AN/SRQ-4 system. The equipment shall provide the operator with the capability to monitor the selected acoustic channel in the supplied headset.

3.5.2.3 Two way voice feature. The equipment shall provide the capability for up and down link voice communications between the equipment aircraft equipment operator. The test set accessory kit shall include a head set boom microphone, headset and extension cable which are compatible with the SH-60B Sensor Operator Station. The received downlink voice level shall be controllable by means of volumn located on the equipment front panel.

3.5.2.4 Fixed computer message transfer. The equipment shall have the capability to selectively transmit any one of six fixed computer messages uplink. The transmission of the fixed message shall be selectable between automatic and manual transmission. The automatic transmission shall occur at a rate that is in consonance with the normal computer message traffic rate. The manual mode shall allow a single tranmission of the designated fixed computer message via the front panel. The fixed computer message transfer will be approved by the procuring activity prior to implementation. (See 6.2.2)

3.5.2.5 Radar. The test set shall verify the presence of a standard APS-124 Radar test pattern as defined below:

a. Verify presence of radar control bits

b. Verify incrementing bearing.

c. Display; mode, range, video type, and radar data confidence bit.

d. Verify presence of radar test pattern video as specified for the Signal Data Convertor, OU-103A.

e. The 1555 bit radar frame and a syncronization signal source shall be supplied at the equipment front panel on a test jack to permit operator viewing using an oscilloscope.

3.5.2.6 NAVSYNCH. The equipment shall exchange with the LAMPS MK III Airborne System, sufficient computer message traffic to satisfy the requirements for system NAVSYNCH.

3.5.2.7 R.F. transmission quality. The equipment shall provide the capability to verify the quality transmissions as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

a. The equipment shall provide a means of attenuating the uplink signal while simultaneously evaluating and displaying the Airborne Data Terminal bit error rate. The equipment transmitter output shall be attenuated with a resolution to one dB. The attenuator shall be adjustable from the front panel and shall have a range of zero through 99 dB. The error count shall be displayed on the equipment front panel by means of a five digit counter with a maximum counting capacity of 99,999 and having an operating speed compatable with the equipment data rate. Error rates shall be accumulated and displayed in at least 1, 10 and 100 second intervals. The counter shall also have the capacity to count continuously. The counter shall indicate to the operator when an overflow condition occurs and shall have the capability to be manually reset. The counter shall have a means of indicating to the operator that a counting interval is in progress. This capability shall permit evaluation of airborne receiver sensitivity.

b. The test set shall provide an independent means of attenuating the downlink signal while simultaneously evaluating received bit error rate. The attenuator shall be the same type as that used to attenuate the uplink signal. This capability shall permit evaluation of airborne transmitter power and modulation capability. An independent counter having the same characteristics as described above shall be provided to display downlink bit error rate.

3.5.2.8 Operating frequency and frequency stability. The operating frequency and the frequency stability shall be as specified for the AN/ARQ-44.

3.5.2.9 Up and down link sync and data quality indications. The equipment shall provide the following sync and data quality indications to the operator.

a. Up link Sync - derived from downlink overhead.

b. Downlink Sync - derived from local sync lock.

c. Data Valid - as derived in the AN/SRQ-4.

3.5.3 Accessory material. The following items shall be provided by the contractor as acessories to the equipment.

a. Equipment operating instructions including a directive of AOP to SPOP computer messages (See 3.5.2.1).

b. KG-45 jumper cables for use in the aircraft

c. Printed Circuit Card Extender(s)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

d. Module Assembly Extender Cable(s)

e. AN/ARQ-44 ATE Connector Interface Cable

f. Primary RF Interface Cable (20 foot RG 393 or equivalent)

g. Baseband Downlink Interface Cable (20 foot RG 62 or equivalent)

h. Baseband Uplink Interface Cable (20 foot RG 62 or equivalent)

j. Acoustic output to helicopeter interface (22 foot twisted/shielded)

k. 60 Hz power cord

m. 400 Hz power cord compatible with the SH-60B utility outlet

n. Six foot length(s) RG 58 with BNC Connectors (as required)

p. Headset with boom microphone, compatible with the SH-60B Communications system (See 3.5.2.3)

r. Grounding Cable (See 3.3.6.7)

3.5.4 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Design Approval Tests (See 4.2)

b. Reliability Development Test (See 4.3)

c. First Article Inspections (See 4.4)

d. Initial Production Tests (See 4.5)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

e. Acceptance Tests (See 4.6)

f. Maintainability Demonstration Tests (See 4.7)

4.2 Design approval tests. The contractor shall conduct Design Approval Tests. These tests shall consist of Environmental Tests in to conform to MIL-T-5422, Electromagnetic Compatibility tests to conform to MIL-STD-462 as ammended herein and functional tests using approved test procedures. (See 6.2.2)

4.3 Reliability development test. A Reliability Development Test (RDT) to conform to MIL-STD-2068 shall be conducted. Test Level E of MIL-STD-781 shall be used except that the minimum temperature shall be -10°C and vibration shall be 10 minutes of each operating hour. A minimum of two equipments shall be put on test for a combined operating time of 1800 hours. A maximum time of 1000 hours shall be not exceeded on any one unit. Acceptance shall be upon procuring activity approval of corrective action for all failures. Changes resulting from Design Approval Tests shall be implemented in the reliability test equipments in a timely and efficient manner during the test.

4.3.1 Redesign during reliability development test. The contractor shall develop and implement corrective action to prevent the recurrence of all failures appearing during the test. In some cases, the logical and effective corrective action will be to redesign portions of the circuitry of the equipment. In addition to this corrective action, the contractor is encouraged to mount an intensive effort to redesign the equipment for maximum simplicity prior to the start and during the early phase of the Reliability Development Test. This redesign for maximum simplicity has the greatest potential for improving the inherent reliability of the equipment. However, any changes to the equipment design are subject to verification by repeating all or portions of the environmental or other Design Approval Tests.

4.3.2 Corrective action. Corrective action shall conform to MIL-STD-781 except that corrective action shall not await a reject but rather on the occasion of a failure (any operation outside specified limits). The procuring activity shall be notified thereof immediately and a proposed plan for corrective action shall be developed promptly. (See 6.2.2). Effective corrective action for all failures shall be developed and implemented to prevent the recurrence of the failures. Formal engineering change proposal procedures are not required. In all cases, the analysis of failures and the corrective actions resulting therefrom shall be documented. (See 6.2.2)

4.3.3 Classification of failures. Failures shall be classified in accordance with MIL-STD-781 and MIL-STD-2074 except that reclassification of failures shall not apply.

4.4 First article inspections. First article inspections shall be conducted on an equipment representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the approved test procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

4.4.1 First article inspection data. The contractor shall make all data collected in conducting these tests available to the procuring activity for review and acceptance. (See 6.2.2)

4.4.2 Scope of tests. First article inspections shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article inspection shall include environmental tests in accordance with the procedures of MIL-T-5422 and an Electromagnetic Compatibility Test in accordance with MIL-STD-461 and MIL-STD-462 as amended herein.

4.4.3 First article approval. Approval of the first article will be by the procuring activity upon satisfactory completion of all tests.

4.4.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equal to the approved first article. Each equipment shall be capable of successfully passing the same tests as imposed on the first article. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.4.5 Electromagnetic compatibility tests. An electromagnetic compatibility test shall be conducted on the equipment to conform to MIL-STD-462 Notice 1 and 2 for Class A1 equipment.

4.5 Initial production tests. One of the first ten production equipments may be selected and sent at the contractor's expense to a designated Government laboratory for test. This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Intitial Production Tests. The first article shall not be selected for this test.

4.5.1 Scope of tests. The equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved first article in design, construction, workmanship, performance and quality and that it meets all applicable requirements of this specification and the contract.

4.5.2 Accessory material. In addition to the complete equipment submitted for initial Production Tests, the contractor shall also submit such accessory material and include data necessary to test the equipment.

4.5.3 Initial production sample approval. Approval of the Initial Production sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during the test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.6 Acceptance test. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Representative. The contractor shall prepare test reports showing quantitative results for all acceptance tests. (See 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests (See 4.6.1)

b. Sampling Tests (See 4.6.2)

c. Reliability Assurance Tests (See 4.6.3)

d. Special Tests (See 4.6.4)

4.6.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests.

a. Examination of Product (See 4.6.1.1)

b. Operational Test (See 4.6.1.2)

c. Manufacturing Run in Test (See 4.6.1.3)

4.6.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.6.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified equipment operation.

4.6.1.3 Manufacturing run-in test. As a minimum, each equipment shall be subjected to a manufacturing run-in test consisting of the following environments:

a. Random vibration.

b. Thermal cycling.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the current version before use.

These environments may be applied sequentially in the order listed. Test levels, durations and tolernaces shall conform to MIL-STD-781 for optional nonstatistical production reliability test with sequential environments and as specified in 4.6.1.3.1 and 4.6.1.3.2. Following random vibration, each equipment produced shall be tested for a minimum period of 40 operating hours in accordance with the conditions specified herein. All failures shall be repaired and the test continued except that the last ten hours shall be failure free. A failure will be defined as any out-of-specification performance or any condition which requires repair of the equipment under test. Adjustment of controls normally accessible to an operator will be permitted.

4.6.1.3.1 Random vibration. Testing of more than one axis may be performed. If this option is authorized, the testing shall be in accordance with the test procedure accepted by the procuring activity.

4.6.1.3.2 Thermal cycling. Each cycle shall consist of lowering the chamber temperature to -10°C, with equipment nonoperating until the equipment temperature stabilizes. During the cold period the equipment shall be turned on and tested, then the chamber temperature shall be raised to +55°C. The equipment shall be tested at the high temperature.

4.6.2 Sampling tests. Equipment selected for sampling tests shall first have passed the individual tests. Equipments will be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1*
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof	

* One out of first ten need not be selected and tested if initial production testsa are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.6.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

c. Manufacturing run-in test specified in 4.6.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However each failure should be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment.

4.6.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failures shall conform to MIL-STD-781 and MIL-STD-2074.

4.6.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or order, all equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Test". The number of equipments to be used shall be not greater than five. For the Qualification Phase, the temperature shall be not less than -10°C, and vibration shall be for ten minutes of every hour. The accept- reject criteria for Test Plan III of MIL-STD-781 shall apply.

4.6.3.2 All equipment test. Each equipment produced shall be subjected to an all equipment test consisting of the following environment:

a. Random vibration.

b. Thermal cycling.

These environments may be applied sequentially in the order listed. Test levels, durations and tolerances shall conform with MIL-STD-781 for optional nonstatistical production acceptance reliability test and as specified in 4.6.3.2.1 and 4.6.3.2.2. Following random vibration, each equipment produced shall be tested for not less than of 50 operating hours in accordance with the conditions specified herein. All failures shall be repaired and the test continued except that the last ten hours of operation shall be failure free. A failure shall be defined as any out-of-specification performance or any condition which requires repair of the equipment under test. Adjustment of the controls normally accessible to an operator will be permitted.

4.6.3.2.1 *Random vibration*. Testing of more than one axis may be performed. If this option is authorized, the testing shall be in accordance with the test procedure approved by the procuring activity.

4.6.3.2.2 Thermal cycling. Each cycle shall consist of lowering the chamber temperature to -10°C, with equipment nonoperating until the equipment temperature stabilizes. During the cold period the equipment shall be turned on and tested, then the chamber temperature shall be raised to +55°C. The equipment shall be tested at the high temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

4.6.3.2.3 Reject-accept criteria. The accept-reject criteria shall conform to the requirements of the all-equipment production reliability acceptance test plan of MIL-STD-781. The lower test MTBF (θ_1) shall be as specified in 3.3.2.4 herein. During the all-equipment test, each equipment shall be exercised and tested in accordance with the approved test procedure. Should a failure occur, the equipment shall be repaired, proper operation verified and the test continued commencing at the beginning of the cycle in which the failure occurred. Corrective action shall be taken, should failures occur, as described in 4.3.2. Records shall be kept of all failures. (See 6.2.2). To determine whether a reject condition has been reached at any time during the contract, the operating test hours and the failures therein shall be totaled and the results compared with the all-equipment production reliability acceptance test plan of MIL-STD-781. At the conclusion of each month, the test results shall be prepared for the procuring activity. Any time that the current totals of test hours and test failures show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment any time that a reject situation exists, pending a review of the Contractor's efforts to improve the equipment. Any corrective actions made shall be incorporated in all units at the contractor's expense. All equipment shall be subjected to and shall pass individual tests prior to the all-equipment test.

4.6.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedure to be prepared by the contractor and accepted by the procuring activity prior to the beginning of the Qualification Test Phase of the all equipment tests.

4.6.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.6.4.1 Special test schedule. Selection of equipment for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other informaton indicate that additional tests are required. (This will be determined by the procuring activity).

4.6.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests. (See 6.2.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

4.6.5 Equipment failure. Should a failure occur during either the sampling reliability assurance, or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare for the procuring activities acceptance, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests or documented justification where no corrective action had been proposed. (See 6.2.2)

d. Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been corrected.

4.7 Maintainability demonstration tests. A maintainability demonstration for intermediate Level requirements shall be conducted to conform to MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B of MIL-STD-471 shall be used.

4.8 Test procedures. The procedures used for conducting all test required by Section 4 shall be prepared by the contractor and be made available to the procuring activity for review and acceptance. (See 6.2.2). The right is reserved by the procuring activity or the Government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.9 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The equipment described by this specification is intended for functional and performance testing of an airborne digital data terminal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of testing. (See Section 4)

c. Corrective action on previously accepted equipment, if required. (See 4.3.2).

c. Selection of applicable levels of packaging or packing. (See 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR SUP 52.227.7029 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

PARAGRAPH NO.	DATA REQUIREMENTS TITLE	APPLICABLE DID	OPTION
3.2.1	Nonstandard Part Approval Requests	DI-E-7028A	
3.2.1	Data, Microelectronic Devices and Assemblies	UDI-E-2133'	
3.2.4.2	Report, Reliability Stress Analysis	UDI-R-21423	
3.3.2	Reliability Program Plan	DI-R-7079	
3.3.2.6.1	Report, Design	DI-R-24039A	
3.3.2.6.1.1	Failure Modes Effects and Critically Analysis Report	DI-R-7085A	
3.3.2.6.2	System/Equipment Failure Analysis and Corrective Action Report	DI-R-5283	
3.3.2.6.2.1	Reports, Test	DI-T-2072	
3.3.4	Drawing Requirement for Rack or Console	UDI-E-21339	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

PARAGRAPH NO.	DATA REQUIREMENTS TITLE	APPLICABLE DID	OPTION
3.3.7.2	Plan Maintainability Program	DI-R-2127	
3.3.7.4.8	Nonstandard Part Approval Requests	DI-E-7028A	
3.3.8	Nomenclature and Identification Assignment and Approval Request	DI-E-21336B	
3.3.17	System Safety Program Plan	DI-H-7047A	
3.5.2.4	Interface Design Specification	DI-E-2135	
4.2	Design Approval Test Procedures	UDI-T-21347	
4.3	Procedures, Reliability Test	DI-R-7035	
4.3.2	Product Assurance, Failure Analysis and Corrective Action Reporting Plan	DI-R-1704	
4.3.2	System/Equipment Failure Analysis and Corrective Action Report	DI-R-5283	
4.4.1	Report First Article (Pre-production Test)	UDI-T-21349	
4.6	Reports, Reliability Test and Demonstration	DI-R-7034	
4.6.3.2.3	Report, Failure	UDI-R-21141A	
4.6.3.3	Procedures, Reliability Tests	DI-R-7035	
4.6.5	Product Assurance, Failure Analysis and Corrective Action Reporting Plan	DI-R-1704	
4.8	Procedures, Reliability Tests	DI-T-21347	

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publication and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first 10 production items. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the documents first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

6.5 Standard sample. Standard sample requirements should be in accordance with 4.5.3, first article approval, of this specification.

6.6 Associate Equipment. The test set should operate with the following associated equipment:

a.	AN/ARM-53B	VHF Sonobuoy Signal Generator
b.	HP-200 CD	Audio Oscillator
c.	Oscilloscope	Delayed Sweep, Externally Triggered 10 MHZ minimum frequency response
d.	AN/ARQ-44	Airborne Data Terminal
e.	AN/UYS-1	Spectrum Analyzer

Preparing activity
Navy-AS
(Project 7035-N008)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85732(AS)	MILITARY SPECIFICATION TEST SET, AIRBORNE DATA TERMINAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.